Chapter 11 — Errors and Debugging

Every program fails sometimes: users type letters into number fields, networks drop, files go missing, and programmers make mistakes. JavaScript reports problems with **error objects** that are **thrown** and can be **caught**. This chapter shows how errors work, how to design your own, and how to find bugs quickly with the console, stack traces and real debuggers — in the browser and in Node.

What you'll learn

1. The built-in error types, what each one means, and what every error object contains.
2. `throw`, `try` / `catch` / `finally` — exact control flow, rethrowing, and the traps.
3. Custom error classes and a clean error-handling layer.
4. Error chaining with `cause` (ES2022) and `AggregateError`.
5. Errors in asynchronous code (the preview; the full story is in Chapter 12).
6. Guaranteed cleanup with `finally` and the new `using` declaration.
7. The console toolbox beyond `console.log`.
8. Reading stack traces; source maps.
9. Debugging with browser DevTools and the Node inspector; a systematic debugging method.
10. A gallery of the most common runtime errors, with their exact messages and causes.

Before you start. Custom errors use classes and `extends` (Chapter 9). Type coercion, which causes many silent bugs, is in Chapter 2.

All runnable code is in `code/`. Outputs are real (Node 24, V8 engine). Error **messages** are not standardized — Firefox and Safari word them differently — but error **types** are.

11.1 Errors and the built-in error types

What it is

An **error** (also called an **exception**) is an object that describes something that went wrong. When code **throws** an error, normal execution stops immediately, and the engine unwinds the call stack looking for a `catch` that will handle it. If none is found, the error is **uncaught**: the browser logs it to the console (and the rest of that script task stops), and Node prints it and exits the process with a non-zero exit code.

Analogy: a fire alarm. Someone pulls it (`throw`); everyone stops what they're doing; the alarm travels up floor by floor until someone responsible (`catch`) deals with it. If nobody does, the building is evacuated (the program crashes).

Terms:

- **Runtime error** — thrown while code runs (`TypeError`, `RangeError`, …).
- **Parse-time (syntax) error** — the engine can't even read the code; *nothing* in that script/module runs.
- **Call stack** — the list of functions currently running, newest on top (Chapter 12 shows it in the event loop).

The built-in types

Type	Thrown when…	Typical example
`Error`	Generic base type — use for your own errors or subclass it	`throw new Error('config missing')`
`TypeError`	A value is the wrong type for an operation	`null.length`, calling a non-function, assigning to a `const`
`ReferenceError`	A name doesn't exist, or is used in its TDZ	typo in a variable name; `let` used before its line
`SyntaxError`	Code (or JSON) can't be parsed	`JSON.parse('{bad}')`, `new Function('1 +')`
`RangeError`	A number is outside the allowed range	`new Array(-1)`, `toFixed(101)`, stack overflow
`URIError`	Malformed URI component	`decodeURIComponent('%')`
`EvalError`	Not thrown by modern engines; exists for backward compatibility	–
`AggregateError` (ES2021)	Several errors need reporting together	`Promise.any` when every promise rejects
`SuppressedError` (ES2027)	An error happened while disposing after another error	`using` declarations (§11.6)

All of them inherit from `Error` (`TypeError.prototype.[[Prototype]] === Error.prototype`), so `err instanceof Error` is `true` for every built-in error. Environments add their own: browsers have `DOMException` (e.g., `AbortError`, `NotAllowedError`), Node adds `code` properties like `'ENOENT'` to system errors.

`code/01_error_types.mjs` triggers each one for real:

```
const cases = {
  TypeError:      () => null.length,
  ReferenceError: () => notDeclaredAnywhere + 1,
  SyntaxError:    () => JSON.parse('{ bad json }'),
  RangeError:     () => new Array(-1),
  URIError:       () => decodeURIComponent('%'),
  EvalError:      () => { throw new EvalError('never thrown by modern engines — only kept for compatibility');
},
  AggregateError: () => { throw new AggregateError([new Error('a'), new TypeError('b')], 'several things
failed'); },
};

for (const fn of Object.values(cases)) {
  try {
    fn();
  } catch (err) {
    console.log(`${err.name.padEnd(15)} | instanceof Error: ${err instanceof Error} | ${err.message}`);
  }
}

const e = new TypeError('price must be a number', { cause: { got: 'ten' } });
console.log('\nname   :', e.name);
console.log('message:', e.message);
console.log('cause  :', e.cause);
console.log('stack  :', e.stack.split('\n')[0], '... (+', e.stack.split('\n').length - 1, 'frame lines)');
console.log('String :', String(e));

console.log('\nproto chain:', Object.getPrototypeOf(TypeError) === Error,
Object.getPrototypeOf(RangeError.prototype) === Error.prototype);

try {
  await Promise.any([Promise.reject(new Error('mirror 1 down')), Promise.reject(new Error('mirror 2 down'))]);
} catch (agg) {
  console.log('\nPromise.any ->', agg.name, '|', agg.message, '|', agg.errors.map((x) => x.message));
}

console.log('\nError.isError:', Error.isError(e), Error.isError({ name: 'Error', message: 'fake' }));

try {
  new Function('return 1 +');   // parsing happens here
} catch (err) {
  console.log('parse-time  :', err.name, '-', err.message);
}
```

Output:

```
TypeError       | instanceof Error: true | Cannot read properties of null (reading 'length')
ReferenceError  | instanceof Error: true | notDeclaredAnywhere is not defined
SyntaxError     | instanceof Error: true | Expected property name or '}' in JSON at position 2 (line 1 column
3)
RangeError      | instanceof Error: true | Invalid array length
URIError        | instanceof Error: true | URI malformed
EvalError       | instanceof Error: true | never thrown by modern engines — only kept for compatibility
AggregateError  | instanceof Error: true | several things failed

name    : TypeError
message: price must be a number
cause   : { got: 'ten' }
stack   : TypeError: price must be a number ... (+ 4 frame lines)
String : TypeError: price must be a number

proto chain: true true

Promise.any -> AggregateError | All promises were rejected | [ 'mirror 1 down', 'mirror 2 down' ]

Error.isError: true false
parse-time  : SyntaxError - Unexpected token '}'
```

Anatomy of an error object

Property	Standard?	Meaning
`name`	✓ (inherited from the prototype)	The type, e.g. `'TypeError'`
`message`	✓	Human-readable description
`cause`	✓ ES2022	The underlying reason, passed as `new Error(msg, { cause })` (§11.4)
`errors`	✓ on `AggregateError`	Array of the individual errors
`stack`	de-facto (every engine; a standard is proposed)	Text of the call stack at creation time
`code`	Node / libraries	Machine-readable code, e.g. `'ENOENT'`, `'ERR_MODULE_NOT_FOUND'`

`String(err)` (and `err.toString()`) gives `"Name: message"`. The stack is captured when the error object is **created** (with `new`), not when it's thrown.

`Error.isError(value)` (ES2026; available in Node 24 and recent browser versions — check support if you target older ones) is the reliable check for "is this a genuine error object?" — unlike `instanceof Error` it works for errors from other realms (iframes, `vm` contexts) and can't be fooled by an object with a hand-made prototype.

Self-check

1. `undefinedFn()` where `undefinedFn` was never declared — which error? → `ReferenceError` (name not found). If it's declared but holds `undefined`, it's a `TypeError` (not a function).
2. Which property holds the underlying error in a wrapped error? → `cause`.
3. When is a `SyntaxError` in your source file reported? → When the file is parsed, before any of it runs.

TL;DR

Errors are objects with `name`, `message`, `stack` (and optional `cause`). Know the core types: `TypeError` (wrong type), `ReferenceError` (unknown name/TDZ), `SyntaxError` (unparseable), `RangeError` (out of range), `AggregateError` (many).

11.2 `throw` , `try` , `catch` , `finally`

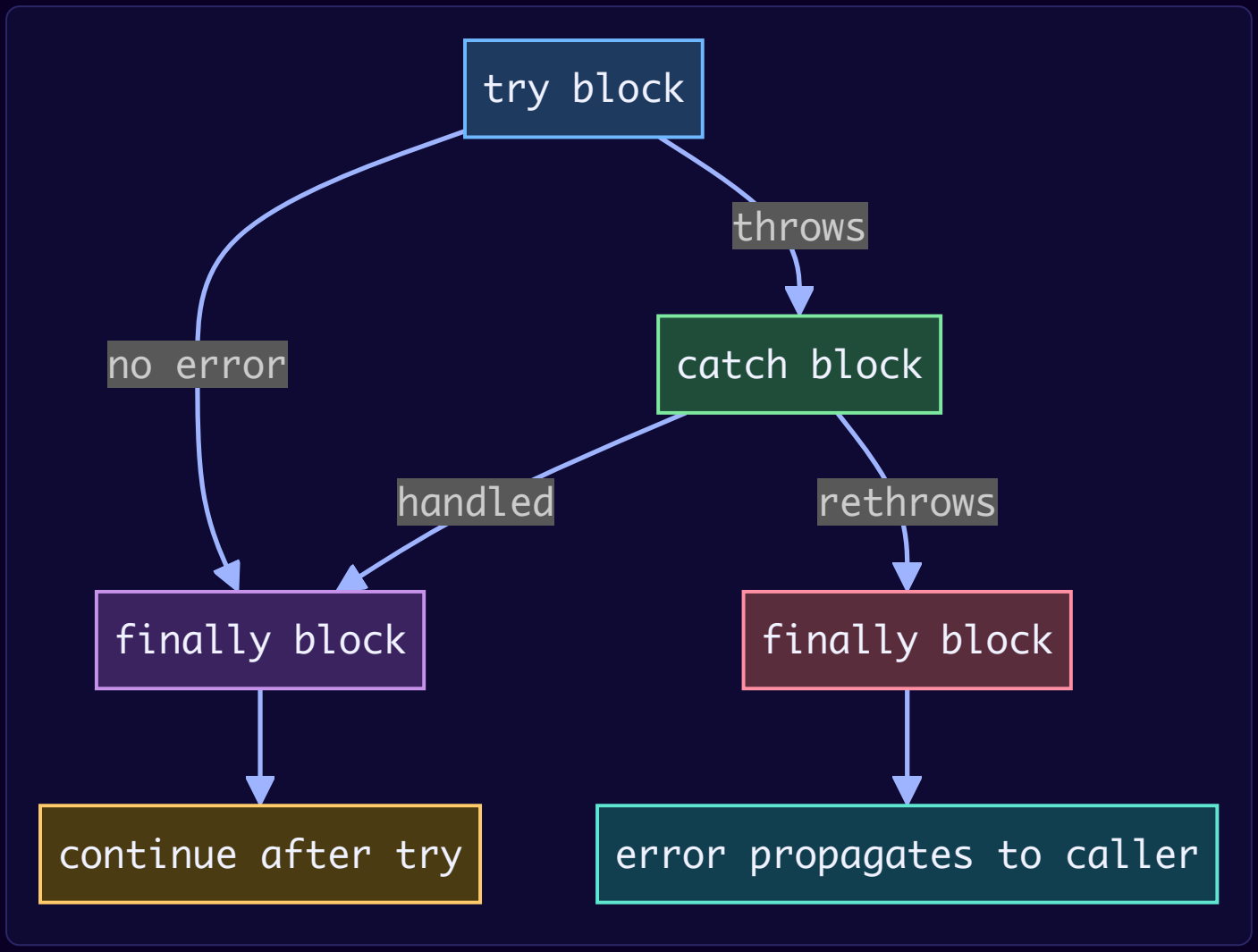
What it is

- `throw expression` – raises an error; execution jumps to the nearest enclosing `catch`.
- `try { … }` – code that might throw.
- `catch (err) { … }` – runs only if something in `try` threw; `err` is the thrown value.
- `finally { … }` – always runs after `try` / `catch`, whether the block succeeded, threw, or `return`ed. Used for cleanup.

A `try` must have a `catch`, a `finally`, or both.

How it works

1. Run the `try` block.
2. If it throws, abandon the rest of `try`, and run `catch` with the error.
3. Then run `finally` (always).
4. If `catch` itself throws (or rethrows), `finally` still runs, then the new error propagates outward.
5. If nothing catches it anywhere up the stack, it's an uncaught error.



`code/02_try_catch_finally.mjs` :

```
function parseAge(text) {
  const log = [];
  try {
    log.push('try start');
    const age = Number.parseInt(text, 10);
    if (Number.isNaN(age)) throw new TypeError(`not a number: "${text}"`);
    if (age < 0) throw new RangeError(`negative age: ${age}`);
    log.push('try end');
    return { age, log };
  } catch (err) {
    log.push(`catch ${err.name}`);
    return { age: null, log };
  } finally {
    log.push('finally');       // runs on success, on error, and even after return
  }
}
for (const input of ['42', 'abc', '-3']) {
  const { age, log } = parseAge(input);
  console.log(input.padEnd(4), '->', String(age).padEnd(5), log.join(' > '));
}

// finally's return OVERRIDES the try's return (and swallows errors) — avoid!
function sneaky() {
  try { throw new Error('lost forever'); }
  finally { return 'finally wins'; }  // eslint: no-unsafe-finally
}
console.log('\nsneaky():', sneaky());

// Optional catch binding (ES2019): omit (err) when you don't need it
function isValidJson(s) {
  try { JSON.parse(s); return true; } catch { return false; }
}
console.log('isValidJson:', isValidJson('{"a":1}'), isValidJson('{a:1}'));

// Catch only what you understand; rethrow the rest
function loadUser(raw) {
  try {
    return JSON.parse(raw).user.name;
  } catch (err) {
    if (err instanceof SyntaxError) return '(invalid JSON — using guest)';
    throw err;                     // a TypeError here means a real bug: don't hide it
  }
}
console.log('rethrow 1  :', loadUser('{oops'));
try { loadUser('{"nouser":true}'); } catch (err) { console.log('rethrow 2  :', err.name, '-', err.message); }

// You can throw anything — but only Error objects carry a stack
for (const thrown of ['a string', 42, { code: 'E_BAD' }, new Error('real error')]) {
  try { throw thrown; } catch (err) {
    console.log('threw', typeof err === 'object' ? err.constructor.name : typeof err, '| has stack:', typeof
err?.stack === 'string');
  }
}

// try/catch is synchronous: it cannot catch an error thrown later by a callback
try {
  setTimeout(() => {
    try { throw new Error('from timer'); } catch (err) { console.log('\ntimer error caught INSIDE the
callback:', err.message); }
  }, 0);
} catch {
  console.log('never printed');
}
console.log('\n(outer try block already finished)');
```

Output:

```
42   -> 42    try start > try end > finally
abc  -> null  try start > catch TypeError > finally
-3   -> null  try start > catch RangeError > finally

sneaky(): finally wins
isValidJson: true false
rethrow 1  : (invalid JSON — using guest)
rethrow 2  : TypeError - Cannot read properties of undefined (reading 'name')
threw string | has stack: false
threw number | has stack: false
threw Object | has stack: false
threw Error | has stack: true

(outer try block already finished)

timer error caught INSIDE the callback: from timer
```

What to notice:

- `finally` ran in all three cases — even though `try` and `catch` both `return`. The return value is computed first, then `finally` runs, then the function actually returns.
- A `return` (or `throw`) *inside* `finally` replaces whatever `try`/`catch` was doing — the error `'lost forever'` vanished. ESLint's `no-unsafe-finally` rule flags this.
- **Catch narrowly, rethrow the rest.** `loadUser` expects bad JSON (a `SyntaxError`) and recovers; a `TypeError` means the data shape is wrong — a real bug that should surface.
- `catch` receives *whatever* was thrown. Throwing strings or plain objects loses the stack trace and breaks `instanceof Error` checks. **Always throw `Error` objects** (or subclasses).
- The `catch` variable is block-scoped to the `catch` block.
- `try/catch` only covers code that runs *synchronously inside it*. The timer callback ran later, on a fresh call stack, after the outer `try` was long gone.

When to use try/catch — and when not

Use it	Don't use it
At boundaries where failure is expected: parsing input, `JSON.parse`, network/file I/O, third-party code	To replace simple checks — `if (user?.name)` is cheaper and clearer than catching a `TypeError`
In a top-level handler that turns errors into a response/UI message and logs them	Wrapping every function "just in case" — it hides bugs
To add context and rethrow (§11.4)	To silently swallow errors: `catch {}` with no comment or handling

Performance note: in modern engines a `try` block that doesn't throw costs essentially nothing; *throwing* is relatively expensive (capturing the stack), so don't use exceptions for normal control flow like "item not found in a loop".

In the interview

30-second answer: "`throw` raises an error and unwinds the stack to the nearest `catch`. `finally` always runs — after success, error, or even `return` — so it's for cleanup; but a `return` inside `finally` overrides everything and swallows errors. `try/catch` is synchronous, so it can't catch errors from callbacks that run later — for promises you `await` inside `try`, or use `.catch()`. Always throw `Error` objects so you get a stack trace."

- *Q: What does `(() => { try { return 1 } finally { return 2 } })()` return?* A: `2`.
- *Q: Does `finally` run if `catch` rethrows?* A: Yes, then the error continues outward.
- *Q: Why is `catch (e) {}` dangerous?* A: It hides every error, including programming bugs, making failures silent and hard to trace.

Self-check

1. Can you have `try { } finally { }` without `catch`? → Yes; errors still propagate, but cleanup runs.
2. What is `err` in `catch (err)` after `throw 42`? → The number `42` (no stack).
3. Why didn't the outer `catch` see the timer's error? → The callback ran later, after the `try` finished.

TL;DR

`try` → `catch` on error → `finally` always. Throw `Error` objects, catch narrowly, rethrow what you can't handle, never `return` from `finally`, and remember try/catch is synchronous.

11.3 Custom errors

What it is

A **custom error** is a subclass of `Error` for a specific failure in *your* domain: `ValidationError`, `NotFoundError`, `PaymentDeclinedError`. It lets callers react to *kinds* of failure with `instanceof` (or a `code`), and carry extra data (which field was invalid, what HTTP status to return).

How it works

1. `class MyError extends Error`.
2. Call `super(message, options)` so `message`, `stack` and `cause` are set up.
3. Set `this.name` (otherwise it prints as plain `Error`). Using `new.target.name` sets it automatically for every subclass.
4. Add fields (`code`, `status`, `field`…).
5. Handle errors in **one place** (a top-level handler, an Express error middleware, a React error boundary) that maps error types to responses, and treats *unknown* errors as 500s without leaking their details to users.

`code/03_custom_errors.mjs`:

```
class AppError extends Error {
  constructor(message, options = {}) {
    super(message, { cause: options.cause });   // pass cause up to Error
    this.name = new.target.name;                // 'ValidationError', 'NotFoundError', ...
    this.code = options.code ?? 'APP_ERROR';
    this.status = options.status ?? 500;
  }
  toJSON() { return { name: this.name, code: this.code, message: this.message }; }
}

class ValidationError extends AppError {
  constructor(field, message) {
    super(message, { code: 'VALIDATION', status: 400 });
    this.field = field;
  }
}
class NotFoundError extends AppError {
  constructor(what, id) { super(`${what} ${id} not found`, { code: 'NOT_FOUND', status: 404 }); }
}

function placeOrder(order) {
  if (!Number.isInteger(order.qty) || order.qty < 1) throw new ValidationError('qty', 'qty must be a positive
integer');
  if (order.productId !== 'p1') throw new NotFoundError('Product', order.productId);
  return { ok: true };
}

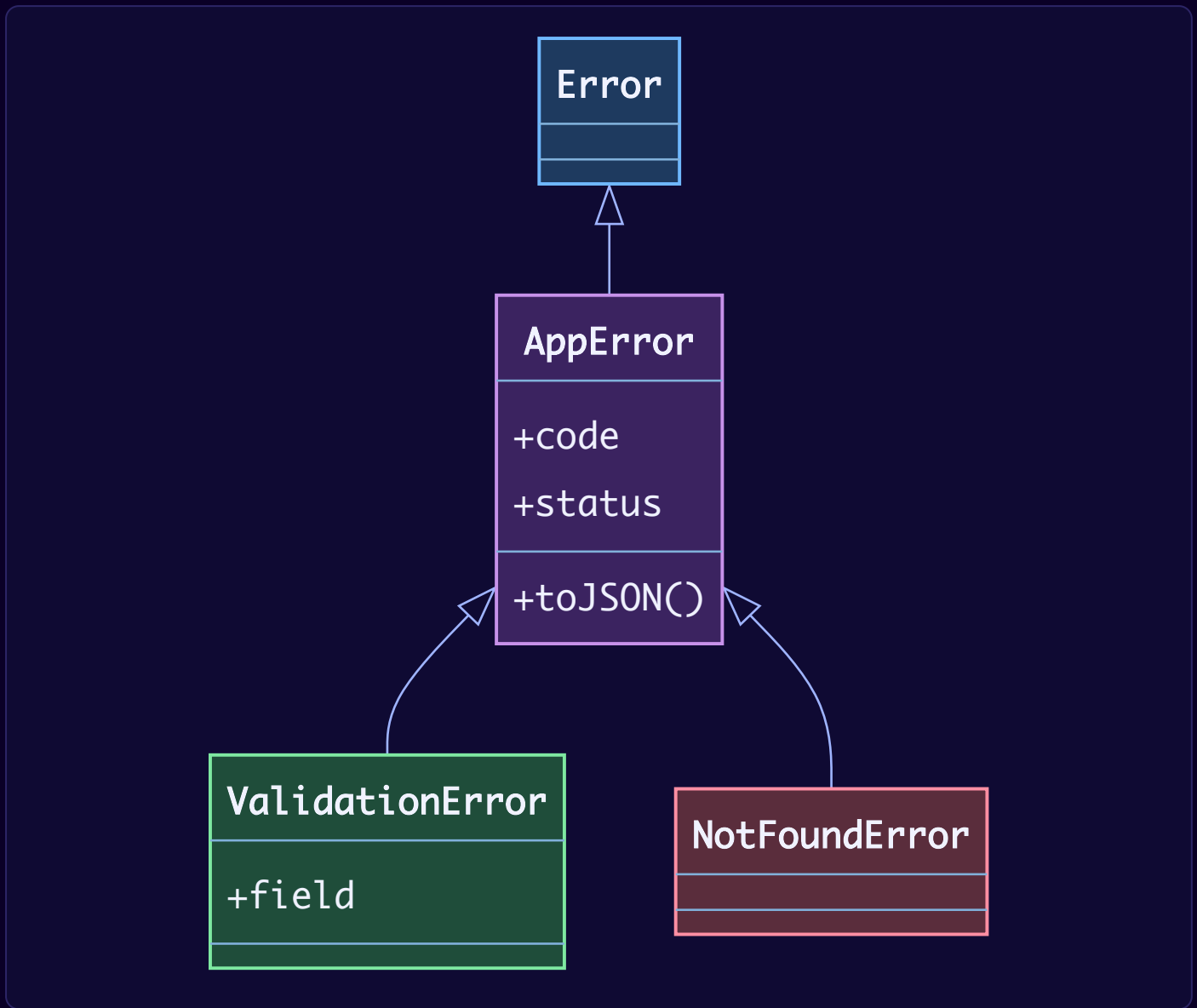
// A single place that turns errors into responses
function handle(order) {
  try {
    return { status: 200, body: placeOrder(order) };
  } catch (err) {
    if (err instanceof AppError) return { status: err.status, body: err.toJSON() };
    return { status: 500, body: { message: 'internal error' } };   // unknown -> don't leak details
  }
}
console.log(handle({ productId: 'p1', qty: 2 }));
console.log(handle({ productId: 'p1', qty: 0 }));
console.log(handle({ productId: 'p9', qty: 1 }));

const v = new ValidationError('email', 'bad email');
console.log('\ninstanceof  :', v instanceof ValidationError, v instanceof AppError, v instanceof Error);
console.log('name/field  :', v.name, v.field);
console.log('String(v)   :', String(v));
console.log('stack line 1:', v.stack.split('\n')[0]);
console.log('JSON        :', JSON.stringify(v));
console.log('plain Error JSON:', JSON.stringify(new Error('x')), '(message/stack are non-enumerable own
props)');
```

Output:

```
{ status: 200, body: { ok: true } }
{
  status: 400,
  body: {
    name: 'ValidationError',
    code: 'VALIDATION',
    message: 'qty must be a positive integer'
  }
}
{
  status: 404,
  body: {
    name: 'NotFoundError',
    code: 'NOT_FOUND',
    message: 'Product p9 not found'
  }
}

instanceof  : true true true
name/field  : ValidationError email
String(v)   : ValidationError: bad email
stack line 1: ValidationError: bad email
JSON        : {"name":"ValidationError","code":"VALIDATION","message":"bad email"}
plain Error JSON: {} (message/stack are non-enumerable own props)
```



Why the stack's first line already says `ValidationError`: V8 builds the `stack` string lazily from the error's `name` the first time it's read – here after `this.name` was set. Don't rely on that everywhere: set `name` in the constructor so all engines agree.

`JSON.stringify(new Error('x'))` gives `{}` because `message` and `stack` are **non-enumerable** own properties, and `name` lives on the prototype. That's why `AppError` defines `toJSON()` – useful for logs and API responses (but never send `stack` to end users).

Error codes vs classes

Approach	Pros	Cons
Subclasses + `instanceof`	Rich, typed, hierarchical (`catch` a whole family)	Breaks across realms/duplicated packages (two copies of the class)
`code` string (`err.code === 'ENOENT'`)	Survives serialization, logs, process boundaries; Node's own convention	No hierarchy; typos go unnoticed
Both (as above)	`instanceof` inside the app, `code` in logs and API bodies	A little more code

Common mistakes

- Forgetting `super(message)` → `ReferenceError` when you touch `this` (Chapter 9).
- Not setting `name` → logs say `Error: …` for everything.
- Leaking internals: sending `err.message` / `err.stack` of *unexpected* errors to clients can reveal SQL, paths or secrets. Log them server-side; send a generic message.
- Over-engineering: dozens of error classes nobody catches differently. Create a class only when some caller will handle it differently.

Real-world examples

Node's system errors carry `code` (`ENOENT`, `EADDRINUSE`, `ECONNREFUSED`); `fetch` rejects with a `TypeError` on network failure and an `AbortError` `DOMException` when aborted; HTTP libraries throw errors with a `status`/`response`; ORMs throw `UniqueConstraintError`-style subclasses.

Self-check

1. Why set `this.name` in a custom error? → So `String(err)`, stacks and logs show the right type.
2. Why does `JSON.stringify(err)` often print `{}`? → `message`/`stack` are non-enumerable; `name` is inherited.
3. What should a top-level handler do with an unknown error? → Log full details internally; return a generic message (e.g., 500).

TL;DR

`class X extends Error`, call `super(message, { cause })`, set `name`, add `code`/data, handle all errors in one place, and never leak internals of unexpected errors.

11.4 Error chaining with `cause`

What it is

When you catch a low-level error and throw a higher-level one ("could not load config"), you want to **add context without losing the original**. Since ES2022, every error constructor accepts an options object: `new Error(message, { cause })`. The original is stored in `err.cause`.

Before `cause`, people either lost the original (bad for debugging) or concatenated messages into unreadable strings.

`code/04_error_cause.mjs`:

```
function readConfigText(name) {
  const err = new Error(`ENOENT: no such file or directory, open '${name}'`);   // pretend fs failed
  err.code = 'ENOENT';
  throw err;
}

function loadConfig(name) {
  try {
    return JSON.parse(readConfigText(name));
  } catch (err) {
    throw new Error(`Could not load config "${name}"`, { cause: err }); // add context, keep original
  }
}

function startServer() {
  try {
    loadConfig('server.json');
  } catch (err) {
    throw new Error('Server failed to start', { cause: err });
  }
}

try {
  startServer();
} catch (err) {
  let depth = 0;
  for (let e = err; e; e = e.cause) {                                    // walk the cause chain
    console.log(`${'  '.repeat(depth++)}${depth > 1 ? 'caused by: ' : ''}${e.message}${e.code ? ` [${e.code}]`
: ''}`);
  }
  console.log('\nroot cause code:', rootCause(err).code);
}

function rootCause(e) { while (e.cause instanceof Error) e = e.cause; return e; }

import { inspect } from 'node:util';
const shown = inspect(new Error('outer', { cause: new Error('inner') }));
console.log('\nutil.inspect shows cause?', shown.includes('[cause]: Error: inner'));
```

Output:

```
Server failed to start
  caused by: Could not load config "server.json"
    caused by: ENOENT: no such file or directory, open 'server.json' [ENOENT]

root cause code: ENOENT

util.inspect shows cause? true
```

Each layer adds *what it was trying to do*; the root cause keeps *what actually failed*. Node's `console.log` / `console.error` and browser DevTools print the `cause` chain automatically.

`AggregateError` is the sibling for "several independent failures": its `errors` array holds each one (e.g., `Promise.any` where every mirror failed, or validation that reports all bad fields at once).

TL;DR

Wrap with `new Error('what I was doing', { cause: err })`; walk `.cause` to the root. Use `AggregateError` for many errors at once.

11.5 Errors in asynchronous code (preview)

The problem it solves

Asynchronous code (timers, network, files) finishes *later*, after the `try` block around the call has already exited. Promises carry errors as **rejections** instead. The event loop and promises are explained in Chapter 12; here are the error-handling rules you need now.

`code/05_async_errors.mjs` :

```
const fetchUser = async (id) => {
  await null;                                   // simulate async work
  if (id !== 1) throw new Error(`user ${id} not found`);
  return { id, name: 'Ada' };
};

// 1) await inside try/catch – the normal way
try {
  await fetchUser(2);
} catch (err) {
  console.log('await + try/catch :', err.message);
}

// 2) .catch() on the promise
await fetchUser(3).catch((err) => console.log('.catch()          :', err.message));

// 3) Promise.allSettled – collect successes and failures together
const results = await Promise.allSettled([fetchUser(1), fetchUser(4)]);
console.log('allSettled        :', results.map((r) => (r.status === 'fulfilled' ? r.value.name :
r.reason.message)));

// 4) Forgetting to await: the try/catch finishes before the promise rejects
process.on('unhandledRejection', (reason) => {
  console.log('unhandledRejection:', reason.message, '(Node would crash here without this handler)');
});
try {
  fetchUser(5);                                 // no await!
} catch {
  console.log('never printed');
}
console.log('after the try     : the rejection has not happened yet');
await new Promise((r) => setTimeout(r, 10));      // let the event loop report it
```

Output:

```
await + try/catch : user 2 not found
.catch()          : user 3 not found
allSettled        : [ 'Ada', 'user 4 not found' ]
after the try     : the rejection has not happened yet
unhandledRejection: user 5 not found (Node would crash here without this handler)
```

Rules:

1. Inside an `async` function, `throw` becomes a rejected promise, and `await` of a rejected promise throws – so `try { await … } catch` works naturally.
2. **Always `await` (or `return`, or `.catch()`) every promise.** A forgotten promise's rejection escapes every `try`. ESLint (with type info, `@typescript-eslint/no-floating-promises`) can flag this.
3. An **unhandled rejection** – a rejected promise nobody handles – crashes Node by default (since Node 15) and logs "Uncaught (in promise)" in browsers.

Last-resort global handlers

Environment	Synchronous uncaught error	Unhandled promise rejection
Browser	`window.addEventListener('error', e => …)`	`window.addEventListener('unhandledrejection', e => …)`
Node	`process.on('uncaughtException', err => …)`	`process.on('unhandledRejection', reason => …)`

Use them to **log and report** (e.g., to an error-tracking service such as Sentry), not to keep going as if nothing happened. After an `uncaughtException`, a Node process is in an unknown state — log, then exit and let a process manager restart it.

TL;DR

`await` inside `try`, or `.catch()` every promise; never leave a promise floating. Global handlers are for logging, not recovery.

11.6 Guaranteed cleanup: `finally` and `using`

What it is

Resources — open files, database connections, locks, timers, subscriptions — must be released even if something fails. `finally` has always done this. **Explicit Resource Management** (the `using` declaration) is a newer feature that makes cleanup automatic: an object with a `[Symbol.dispose]()` method, declared with `using`, is disposed when the enclosing block exits, however it exits. It reached TC39 stage 4 in 2025 and is slated for ES2027; Node 24 and Chromium-based browsers (Chrome 134+) already support it — check your targets before relying on it in browsers. It mirrors C#'s `using`, Python's `with` and Java's try-with-resources.

`code/09_using_and_cleanup.mjs`:

```
function openFile(name) {
  console.log(`open  ${name}`);
  return {
    name,
    read() { if (name === 'bad.txt') throw new Error(`cannot read ${name}`); return `contents of ${name}`; },
    [Symbol.dispose]() { console.log(`close ${name}`); },   // called automatically by `using`
  };
}

// Classic pattern: try/finally
function classic(name) {
  const f = openFile(name);
  try { return f.read(); }
  finally { f[Symbol.dispose](); }
}
console.log('classic ->', classic('a.txt'));

// Same thing with `using`: disposed when the block exits, in reverse order, even on error
function modern(...names) {
  using first = openFile(names[0]);
  using second = openFile(names[1]);
  return [first.read(), second.read()];
}
console.log('modern  ->', modern('a.txt', 'b.txt'));
try {
  modern('c.txt', 'bad.txt');
} catch (err) {
  console.log('modern error ->', err.message, '(both files were still closed)');
}

// If disposal ALSO throws, you get a SuppressedError holding both errors
function doubleFailure() {
  using r = { [Symbol.dispose]() { throw new Error('dispose failed'); } };
  throw new Error('body failed');
}
try { doubleFailure(); } catch (err) {
  console.log(`${err.name}: error="${err.error.message}" suppressed="${err.suppressed.message}"`);
}
```

Output:

```
open  a.txt
close a.txt
classic -> contents of a.txt
open  a.txt
open  b.txt
close b.txt
close a.txt
modern  -> [ 'contents of a.txt', 'contents of b.txt' ]
open  c.txt
open  bad.txt
close bad.txt
close c.txt
modern error -> cannot read bad.txt (both files were still closed)
SuppressedError: error="dispose failed" suppressed="body failed"
```

- Resources are disposed in **reverse** order of declaration (last opened, first closed) — like a stack.
- `await using` does the same with `[Symbol.asyncDispose]()` for async cleanup (closing a DB connection).
- If both the body and the disposal throw, the result is a `SuppressedError`: `.error` is the disposal error, `.suppressed` the one it replaced — nothing is silently lost.

TL;DR

Release resources in `finally`, or with `using` + `[Symbol.dispose]` where supported. Disposal runs in reverse order, even on errors.

11.7 The console toolbox

What it is

`console` is the built-in logging object in browsers and Node. `console.log` is only the start; the other methods save a lot of time when debugging. In the browser they print to the DevTools **Console** panel (with filtering by level); in Node, `log/info/debug/table/dir/group/count/time` write to **stdout** and `warn/error/trace/assert` to **stderr**.

`code/06_console_methods.mjs`:

```
console.log('log   : plain message', { id: 7 }, [1, 2]);
console.info('info  : alias of log in Node');
console.debug('debug : alias of log in Node (hidden by default in browser DevTools "Verbose" level)');
console.log('format: %s is %d years, %o', 'Ada', 36, { lang: 'js' });

const users = [
  { id: 1, name: 'Ada', role: 'admin' },
  { id: 2, name: 'Linus', role: 'dev' },
];
console.table(users);
console.table(users, ['name']);            // choose columns

const deep = { a: { b: { c: { d: { e: 'deep' } } } } };
console.log('default depth:', deep);
console.dir(deep, { depth: null });         // print everything

console.group('group: checkout');
console.log('validate cart');
console.group('payment');
console.log('charge card');
console.groupEnd();
console.groupEnd();

for (const k of ['a', 'b', 'a', 'a']) console.count(k);
console.countReset('a');
console.count('a');

console.assert(1 + 1 === 2, 'not printed: assertion holds');
console.assert(0.1 + 0.2 === 0.3, 'assertion failed: floating point strikes again'); // prints, does NOT throw

const label = 'sum loop';
console.time(label);
let s = 0; for (let i = 0; i < 1e6; i++) s += i;
console.timeEnd(label);                     // prints elapsed time (varies per run)

function inner() { console.trace('trace: how did we get here?'); }
inner();
```

Output (the timing varies per run; the trace's long file paths are shortened here):

```
log   : plain message { id: 7 } [ 1, 2 ]
info  : alias of log in Node
debug : alias of log in Node (hidden by default in browser DevTools "Verbose" level)
format: Ada is 36 years, { lang: 'js' }
┌─────────┬────┬─────────┬─────────┐
│ (index) │ id │ name    │ role    │
├─────────┼────┼─────────┼─────────┤
│ 0       │ 1  │ 'Ada'   │ 'admin' │
│ 1       │ 2  │ 'Linus' │ 'dev'   │
└─────────┴────┴─────────┴─────────┘

┌─────────┬─────────┐
│ (index) │ name    │
├─────────┼─────────┤
│ 0       │ 'Ada'   │
│ 1       │ 'Linus' │
└─────────┴─────────┘
default depth: { a: { b: { c: [Object] } } }
{
  a: {
    b: { c: { d: { e: 'deep' } } }
  }
}
group: checkout
  validate cart
  payment
    charge card
a: 1
b: 1
a: 2
a: 3
a: 1
Assertion failed: assertion failed: floating point strikes again
sum loop: 1.871ms
Trace: trace: how did we get here?
    at inner (file:///…/11-errors-and-debugging/code/06_console_methods.mjs:38:28)
    at file:///…/11-errors-and-debugging/code/06_console_methods.mjs:39:1
    at ModuleJob.run (node:internal/modules/esm/module_job:377:25)
    …
```

Method	Use it for
`log` / `info` / `debug`	General output; in browsers they're separate filter levels
`warn` / `error`	Problems — shown in yellow/red in DevTools, go to stderr in Node, include a stack in browsers
`%s %d %i %f %o %O %c`	Format specifiers (string, integer, float, object; `%c` applies CSS in browsers)
`table(data, cols?)`	Arrays of objects → readable grid
`dir(obj, { depth })`	Full object inspection (Node's `console.log` stops at depth 2 and shows `[Object]`)
`group` / `groupCollapsed` / `groupEnd`	Indent related messages
`count(label)` / `countReset`	"How many times did this run?"
`assert(cond, msg)`	Log only when `cond` is false — **does not throw**
`time` / `timeLog` / `timeEnd`	Quick timing (use `performance.now()` or a profiler for serious measurement)
`trace(msg)`	Print the current stack without throwing

Common mistakes

- **Logging objects in the browser and reading them later**: DevTools shows a *live* reference; when you expand it, you see the object's *current* state, not the state at log time. Log a snapshot: `console.log(structuredClone(obj))` or `JSON.stringify(obj)`.
- Leaving debug logs in production code – use a lint rule (`no-console`) or a logger with levels.
- `console.log('user: ' + user)` → `user: [object Object]`. Pass objects as separate arguments: `console.log('user:', user)`.

TL;DR

Use `table`, `dir`, `group`, `count`, `assert`, `time`, `trace` – and remember browser logs of objects are live references.

11.8 Reading stack traces and source maps

What it is

A **stack trace** lists the chain of function calls active when the error was created – **newest first**. Each line ("frame") shows the function name and the location `file:line:column`. Reading it top-down tells you *where* it failed; reading bottom-up tells you *how you got there*.

`code/07_stack_traces.mjs` (key parts):

```
function validate(cart) { if (cart.items.length === 0) throw new Error('empty cart'); }
function checkout(cart) { validate(cart); }
function handleClick() { checkout({ items: [] }); }

try {
  handleClick();
} catch (err) {
  // shorten absolute file paths so the output is the same on every machine
  const pretty = err.stack.replace(/file:\/\/\S+\//g, '').replace(/\(.*[\\/](\S+?:\d+:\d+)\)/g, '($1)');
  console.log(pretty.split('\n').slice(0, 5).join('\n'));
}
console.log('\nError.stackTraceLimit default:', Error.stackTraceLimit);
// ... Error.captureStackTrace(obj) attaches a .stack to any object
```

Output:

```
Error: empty cart
    at validate (07_stack_traces.mjs:4:62)
    at checkout (07_stack_traces.mjs:5:27)
    at handleClick (07_stack_traces.mjs:6:26)
    at 07_stack_traces.mjs:9:3

Error.stackTraceLimit default: 10
frames mention makeTimeout? true
captureStackTrace on plain object: string 07_stack_traces.mjs
```

How to read it:

1. **Line 1** = `name: message` – *what* happened.
2. **First frame** = where it was thrown (`validate`, line 4, column 62).
3. **Following frames** = callers. The first frame that's in *your* code (not a library or `node:internal`) is usually where to start looking.
4. **Anonymous frames** (`at 07_stack_traces.mjs:9:3`) are top-level code or unnamed functions – naming your functions makes traces easier to read.

V8-specific helpers (Node, Chrome, Edge — not standard): `Error.stackTraceLimit` (frames kept, default 10; raise it when debugging deep stacks, or run Node with `--stack-trace-limit=50`), and `Error.captureStackTrace(obj, fn?)`, which attaches a stack to any object (optionally hiding frames above `fn`). Async stack traces: V8 stitches together frames across `await` points (you'll see `at async functionName`), so `async/await` code has far more useful traces than old callback code.

Source maps

Production JavaScript is usually bundled and minified (Chapter 10), so a raw trace says something like `at e (app.3f9a1c.js:1:48213)` — useless. A **source map** (`app.js.map`) maps each position in the generated file back to the original file, line and column. DevTools applies source maps automatically when it finds a `//# sourceMappingURL=` comment; Node applies them to stack traces with `node --enable-source-maps`. Error-tracking services let you upload maps privately so production traces are readable without publishing your source.

TL;DR

Read the first line for *what*, the first frame in your code for *where*, the rest for *how*. Use source maps for bundled code; name functions for clearer traces.

11.9 Debugging with DevTools and the Node inspector

What it is

A **debugger** lets you **pause** a running program at a chosen line, **inspect** every variable in scope, and **step** through the code one line at a time. It's far faster than adding `console.log`s and re-running when the problem is "why is this value wrong here?". Browsers ship one in **DevTools** (F12, or Cmd+Option+I on macOS); Node exposes the same Chrome DevTools protocol through its **inspector**.

Core concepts

Concept	What it does
Breakpoint	Pauses execution when a line is about to run. Set by clicking a line number (Sources panel in Chrome, Debugger in Firefox) or with a `debugger;` statement (ignored when no debugger is attached).
Conditional breakpoint	Pauses only when an expression is true (`item.qty > 100`) — essential inside loops.
Logpoint	Logs a message without pausing and without editing the code.
Pause on exceptions	Stops at the line that throws (optionally including caught exceptions) — shows the state at the moment of failure.
Step over (F10)	Run the current line, including any functions it calls, and stop at the next line.
Step into (F11)	Enter the function called on this line.
Step out (Shift+F11)	Finish the current function and stop in its caller.
Resume (F8)	Continue until the next breakpoint.
Call Stack pane	The frames (as in a stack trace) — click one to inspect that caller's variables.
Scope pane	Local, closure and global variables for the selected frame (including closures from Chapter 4 and private `#fields`).
Watch expressions	Expressions re-evaluated at every pause.
Console while paused	Evaluate any expression in the paused scope, even change variables.

(Keyboard shortcuts are Chrome DevTools' defaults; VS Code uses F10/F11/Shift+F11 too, with F5 to continue.)

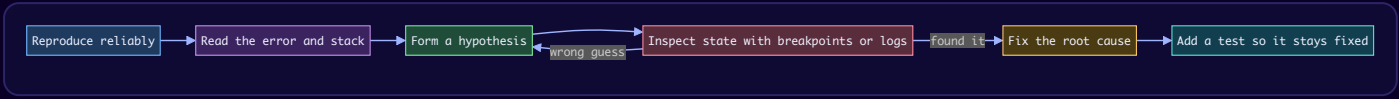
Browser-only extras: **DOM breakpoints** (pause when an element's subtree/attributes change), **event listener breakpoints** (pause on any `click`), **XHR/fetch breakpoints** (pause when a URL is requested), the **Network** panel (see every request, status and response — the first stop for "the API call failed" bugs), **Performance** and **Memory** panels (profiling and heap snapshots for leaks), and an **ignore list** that hides framework/library frames when stepping.

Debugging Node (not run — interactive)

```
node --inspect app.mjs          # start with the inspector on 127.0.0.1:9229
node --inspect-brk app.mjs      # same, but pause on the first line (so you can set breakpoints)
# then open chrome://inspect in Chrome -> "Open dedicated DevTools for Node"
node inspect app.mjs            # built-in terminal debugger: c(ont), n(ext), s(tep), o(ut), repl, bt
node --enable-source-maps dist/app.js   # readable traces for transpiled code
node --watch app.mjs            # restart automatically on save while you fix things
```

In **VS Code**, open a "JavaScript Debug Terminal" and run the program normally — breakpoints set in the editor just work, for Node scripts, test runners and dev servers alike. Never expose `--inspect` on a public network interface: anyone who can connect can run code in your process.

A systematic method



1. **Reproduce** the bug reliably, with the smallest input that triggers it.
2. **Read** the actual error message and stack (really read it).
3. **Hypothesize** what's wrong, then **inspect** the values that would confirm or refute it — breakpoint/`debugger` or targeted, labelled logs.
4. **Narrow down** by halving: is the value already wrong at the midpoint? (`git bisect` does the same across commits.)
5. **Fix the root cause**, not the symptom.
6. **Add a test** that fails without the fix (Chapter 16).

Explaining the code line by line to someone — or to a rubber duck — works surprisingly often, because it forces you to state your assumptions.

Worked example

`code/10_debugging_walkthrough.mjs`: a cart total that comes out wrong.

```
import assert from 'node:assert/strict';

// Data as it arrives from an HTML form or query string: everything is a string.
const cart = [
  { name: 'Book', price: '12.50', qty: '2' },
  { name: 'Pen', price: '1.25', qty: '4' },
];

// BUGGY version
function totalBuggy(items) {
  return items.reduce((sum, item) => sum + item.price * item.qty, '');
}
console.log('symptom       :', totalBuggy(cart), '(expected 30)');

// Inspect the intermediate values — labelled and structured
function totalTraced(items) {
  return items.reduce((sum, item, i) => {
    const next = sum + item.price * item.qty;
    console.log(`  step ${i}:`, { sum, typeofSum: typeof sum, line: item.price * item.qty, next });
    // debugger;   // <- with DevTools / `node inspect` attached, execution would pause here
    return next;
  }, '');
}
console.log('inspect       :');
totalTraced(cart);

// Fix: numeric initial value + explicit conversion at the boundary.
function total(items) {
  return items.reduce((sum, item) => sum + Number(item.price) * Number(item.qty), 0);
}
console.log('fixed         :', total(cart));

assert.equal(total(cart), 30);
assert.equal(total([]), 0);
assert.throws(() => assert.equal(totalBuggy(cart), 30), assert.AssertionError);
console.log('tests         : passed');
```

Output:

```
symptom       : 255 (expected 30)
inspect       :
  step 0: { sum: '', typeofSum: 'string', line: 25, next: '25' }
  step 1: { sum: '25', typeofSum: 'string', line: 5, next: '255' }
fixed         : 30
tests         : passed
```

The inspection shows `typeofSum: 'string'` from the very first step: the initial value `''` made `+` concatenate (`'25' + 5` → `'255'`), while `*` had quietly converted the strings to numbers — a coercion rule from Chapter 2. No error was thrown at all, which is why these bugs are hard: **the most dangerous bugs are wrong values, not exceptions.** Converting input at the boundary (`Number(...)`) and adding tests prevents the whole class of bug. TypeScript would also have flagged `string * string` (Chapter 16).

In the interview

> **30-second answer:** "I reproduce the bug first, read the stack trace, then use breakpoints — often conditional ones or 'pause on exceptions' — to inspect the actual state instead of guessing. I step over/into to follow the flow and check the Network panel for API issues. For Node I use `--inspect` with Chrome DevTools or VS Code's debug terminal. Once fixed, I add a test that reproduces the bug."

- *Q: What's a logpoint?* A: A breakpoint that logs an expression instead of pausing — logging without editing code.

- *Q: How do you debug minified production errors?* A: Source maps, uploaded to the error tracker or loaded in DevTools.
- *Q: What does the `debugger` statement do without DevTools open?* A: Nothing.

TL;DR

Pause, inspect, step: breakpoints (conditional, logpoints, pause-on-exceptions), Scope/Watch/Call Stack panes, Network panel. Node: `--inspect` / `--inspect-brk` + `chrome://inspect` or VS Code. Reproduce → hypothesize → inspect → fix root cause → add a test.

11.10 Common runtime errors — a field guide

`code/08_common_runtime_errors.mjs` triggers twenty everyday mistakes and prints V8's exact messages (what you'll see in Node, Chrome and Edge):

```
const gallery = [
  ['call a non-function', () => { const user = {}; user.getName(); }],
  ['property of undefined', () => { const res = {}; return res.data.items; }],
  ['property of null', () => { const el = null; return el.textContent; }],
  ['set property on undefined', () => { let cfg; cfg.debug = true; }],
  ['undeclared variable', () => totl * 2],
  ['TDZ (use before let)', () => { const f = () => x; f(); let x = 1; }],
  ['reassign const', () => { const n = 1; n = 2; }],
  ['not iterable', () => { const [a] = {}; }],
  ['spread non-iterable', () => [...123]],
  ['JSON.parse(undefined)', () => JSON.parse(undefined)],
  ['JSON.parse(HTML)', () => JSON.parse('<!DOCTYPE html>')],
  ['class without new', () => { class Cart {} Cart(); }],
  ['reduce empty array', () => [].reduce((a, b) => a + b)],
  ['BigInt + Number', () => 1n + 1],
  ['bad array length', () => new Array(2 ** 32)],
  ['toFixed digits', () => (1.5).toFixed(101)],
  ['infinite recursion', () => { const f = () => f(); f(); }],
  ['circular JSON', () => { const a = {}; a.self = a; JSON.stringify(a); }],
  ['frozen object (strict)', () => { const o = Object.freeze({ a: 1 }); o.a = 2; }],
  ['in on a primitive', () => 'x' in 'hello'],
];
for (const [what, fn] of gallery) {
  try { fn(); console.log(`${what.padEnd(26)} (no error)`); }
  catch (e) { console.log(`${what.padEnd(26)} ${e.name}: ${e.message.split('\n')[0]}`); }
}
```

Output:

```
call a non-function        TypeError: user.getName is not a function
property of undefined      TypeError: Cannot read properties of undefined (reading 'items')
property of null           TypeError: Cannot read properties of null (reading 'textContent')
set property on undefined  TypeError: Cannot set properties of undefined (setting 'debug')
undeclared variable        ReferenceError: totl is not defined
TDZ (use before let)       ReferenceError: Cannot access 'x' before initialization
reassign const             TypeError: Assignment to constant variable.
not iterable               TypeError: {} is not iterable
spread non-iterable        TypeError: 123 is not iterable
JSON.parse(undefined)      SyntaxError: "undefined" is not valid JSON
JSON.parse(HTML)           SyntaxError: Unexpected token '<', "<!DOCTYPE html>" is not valid JSON
class without new          TypeError: Class constructor Cart cannot be invoked without 'new'
reduce empty array         TypeError: Reduce of empty array with no initial value
BigInt + Number            TypeError: Cannot mix BigInt and other types, use explicit conversions
bad array length           RangeError: Invalid array length
toFixed digits             RangeError: toFixed() digits argument must be between 0 and 100
infinite recursion         RangeError: Maximum call stack size exceeded
circular JSON              TypeError: Converting circular structure to JSON
frozen object (strict)     TypeError: Cannot assign to read only property 'a' of object '#<Object>'
in on a primitive          TypeError: Cannot use 'in' operator to search for 'x' in hello
```

What each usually means in real code

Message (V8)	Usual real-world cause	Fix
`x is not a function`	Typo in a method name; wrong import (default vs named, Chapter 10); calling a property that holds data; a library version without that method	Check the name and the import; `console.log(typeof x)`
`Cannot read properties of undefined (reading 'y')`	Data not loaded yet (async), wrong shape of an API response, array index out of range, missing `return`	Find *which* value is `undefined` (the part before `.y`); use `?.` and defaults only if absence is legitimate
`Cannot set properties of undefined`	Writing into a nested object that doesn't exist	Create it first, or `obj.cfg ??= {}`
`x is not defined`	Typo, missing import, variable in another scope, browser-only global used in Node (`window`, `document`)	Declare/import it; check the environment
`Cannot access 'x' before initialization`	`let` / `const` / `class` used before its line (TDZ), often via circular imports	Reorder code; break import cycles
`Assignment to constant variable.`	Reassigning a `const` or an imported binding	Use `let`, or mutate instead of reassign
`… is not iterable`	`for…of` / spread / destructuring on a plain object, `undefined`, or a number	Use `Object.entries(obj)`, check for `undefined`
`"undefined" is not valid JSON` / `Unexpected token '<' … is not valid JSON`	Parsing a missing value; the server returned an HTML error page instead of JSON	Check `response.ok` and the content type before `response.json()` (Chapter 14)
`Class constructor X cannot be invoked without 'new'`	Calling a class like a function	Add `new`
`Reduce of empty array with no initial value`	`reduce` on a possibly empty array	Always pass an initial value
`Cannot mix BigInt and other types`	`bigint + number`	Convert explicitly: `BigInt(n)` or `Number(b)`
`Maximum call stack size exceeded`	Infinite or too-deep recursion; a setter that assigns to itself; two functions calling each other	Fix the base case; iterate instead
`Converting circular structure to JSON`	Serializing objects with back-references (DOM nodes, parent links, some library objects)	Pick the fields you need, or use a replacer
`Cannot assign to read only property`	Writing to a frozen object or a getter-only accessor in strict code	Copy instead of mutating: `{ ...obj, a: 2 }`
`Cannot use 'in' operator to search for …`	`in` on a string/number (often an un-parsed JSON string)	Parse first; `in` works on objects only

Other messages worth recognizing: `Unexpected token` / `Unexpected end of input` (syntax: a missing bracket or comma), `await is only valid in async functions and the top level bodies of modules`, `Cannot use import statement outside a module` (Chapter 10), and `Failed to fetch` (network/CORS failure in browsers, a `TypeError` – Chapter 14).

Self-check

1. `Cannot read properties of undefined (reading 'map')` – what is `undefined`? → The value *before* `.map`, not `map` itself.
2. `JSON.parse` fails with `Unexpected token '<'` after a `fetch` – likely cause? → The server returned HTML (an error page or `index.html`), not JSON.

3. Why does `[].reduce((a, b) => a + b)` throw but `[].reduce((a, b) => a + b, 0)` doesn't? → With no initial value, reduce needs at least one element to start from.

TL;DR

Most runtime errors are `TypeError`s from `undefined` / `null` or wrong types. Read the message literally: it names the value and the property. Find where the bad value came from, not just where it exploded.

Chapter 11 TL;DR

- Errors are objects (`name`, `message`, `stack`, optional `cause`); core types: `TypeError`, `ReferenceError`, `SyntaxError`, `RangeError`, `URIError`, `AggregateError`, plus `SuppressedError` for `using`. `Error.isError` (ES2026) is the robust check.
- `try` → `catch` → `finally`: finally always runs; never `return` from it. Catch narrowly, rethrow the rest, always throw `Error` objects. try/catch is synchronous.
- **Custom errors**: `extends Error`, `super(msg, { cause })`, set `name`, add `code`/data, handle centrally, don't leak internals.
- `cause` chains context onto the original error; walk `.cause` to the root.
- **Async**: `await` inside `try` or `.catch()` every promise; global `error` / `unhandledrejection` (browser) and `uncaughtException` / `unhandledRejection` (Node) handlers are for logging.
- **Cleanup**: `finally`, or `using` + `Symbol.dispose` (ES2027, Node 24).
- **Console**: `table`, `dir`, `group`, `count`, `assert`, `time`, `trace`; browser object logs are live.
- **Stack traces**: newest frame first; source maps for bundled code.
- **Debuggers**: breakpoints (conditional, logpoints, pause on exceptions), step over/into/out, Scope/Watch/Call Stack, Network panel; Node `--inspect` + `chrome://inspect` or VS Code.
- **Method**: reproduce → read → hypothesize → inspect → fix root cause → add a test.

Next: the event loop, promises and `async` / `await` – Chapter 12: Asynchronous JavaScript.